

# MAF220 — Numerical Modelling

Computer Lab, Fall 2026

## Exercise 1a:

- Install Python on your computer. Python can be downloaded on MacOS, Windows, and Linux in a free, open-source distribution called Anaconda that makes package management simple. Anaconda can be found at:

<https://www.anaconda.com/>

You may use this or any other distribution. You can also run python in the online environment Google Colab:

<https://colab.research.google.com>

As Colab is less customizable and computational resources more limited it is better for you to have a local python installation.

- Once you have Python running, it's time to write your first Python codes:
  - Write a Python .py file that prints 'Hello World' to the screen and run it.
  - Open Python in interactive mode and make it say 'Hello World'.
  - Open a Jupyter Notebook and do the same.

## Exercise 1b:

If a body of mass  $m$  is in a circular orbit of radius  $r$  around another body of mass  $M$ , and the gravitational pull is the only force acting on it, we know the relation:

$$\frac{mv^2}{r} = \frac{GmM}{r^2} \quad (1)$$

Here,  $v$  is the orbital velocity, and  $G$  is Newton's gravitational constant. Furthermore, we have assumed that  $m \ll M$ . If we also introduce the orbital period  $T$  (the time it takes to do one revolution), we get  $v = \frac{2\pi r}{T}$ , and this expression can be written as:

$$T^2 = \frac{4\pi^2 r^3}{GM} \quad (2)$$

This is Kepler's third law. If we do not assume  $m \ll M$ , this equation still holds if we replace  $M$  by  $M + m$ . If the orbit is an ellipse instead of a circle, we need to replace  $r$  by  $a$ , the semi-major axis of the ellipse.

In astronomy, it is common to use astronomical units instead of regular SI-units. The orbital period would be measured in years (earth years!), the radius of the orbit would be measured in astronomical units (AU, 1 AU =  $149.6 \times 10^6$  km, which is the mean distance from the earth to the sun), and the mass  $M$  would be measured in solar masses ( $M_\odot = 1.99 \times 10^{30}$  kg). Since the earth orbits the sun in  $T = 1$  year at a distance  $r = 1$  AU and the sun has mass  $M_\odot$ , we see from the equation above that in such units Newton's gravitational constant is:

$$G = \frac{4\pi^2 r^3}{MT^2} = \frac{4\pi^2 (\text{AU})^3}{M_\odot \times \text{year}^2} \quad (3)$$

The benefit of using such units is that the factor  $\frac{G}{4\pi^2}$  falls away from the equation above. If we give  $T$  in years and  $M$  in solar masses in the program, we may write the equations for the numerical values as:

$$T^2 = \frac{r^3}{M} \quad [r \text{ in AU}, M \text{ in } M_{\odot}, T \text{ in years}] \quad (4)$$

In the following exercise we therefore use astronomical units.

1. Write a Python function that returns the orbital period  $T$  (in years) when you input the radius  $r$  in AU and the mass  $M$  in solar masses. The function should be of the form:

$$\text{kepler}(r, M) \quad (5)$$

2. Make a plot that compares  $T$  as a function of  $r$  for our solar system with a system that (probably) exists around the star Vega, which has a mass of approximately  $M = 2.0M_{\odot}$ . For this, create a list of  $r$ -values and call your newly created function to get the corresponding orbital period.
3. Save the plot as a PDF file.